

Notes: Berg et al. (RFS, 2020)
On the Rise of FinTechs: Credit Scoring Using Digital Footprints

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1 Big picture

- **Question.** Can very simple online “digital footprint” variables — information left behind when a user visits or registers on a website — predict consumer default as well as traditional credit bureau scores, and do they meaningfully change lending outcomes?
- **Setting.** The paper studies a large German e-commerce firm that ships goods first and invoices customers later. The data cover about 270,000 purchases from October 2015 to December 2016, together with a private credit bureau score and ten easy-to-collect digital-footprint variables.
- **Core challenge.** A useful digital footprint must do more than just repackage information already contained in the credit bureau score or proxy mechanically for time, region, age, gender, or product mix. The paper therefore has to show both predictive power and incremental information.
- **Main conceptual contribution.** The paper studies a type of information that is unusually cheap and scalable. The variables do not require borrowers to upload financial statements, write listing text, reveal social-network links, or provide bank-account data. They are generated almost automatically by ordinary online behavior.
- **Main economic interpretation.** The digital footprint appears to proxy for three broad classes of information:
 - *income / wealth* (for example iOS vs. Android, paid vs. free e-mail providers),
 - *character / self-control* (for example shopping via paid ads or purchasing late at night),
 - *reputation / care* (for example whether the customer’s name appears in the e-mail address, whether the e-mail contains errors, or whether the customer types everything in lowercase).
- **Main empirical design.** The authors compare the discriminatory power of credit bureau scores and digital footprints using logistic default regressions, AUC statistics, out-of-sample tests, future-credit-score forecasting, and a pre/post adoption design around the firm’s introduction of digital-footprint scoring on October 19, 2015.
- **Main baseline result.** For scorable customers, the AUC of the credit bureau score alone is 68.3%, the AUC of the digital footprint alone is 69.6%, and the AUC of the combined model is 73.6%. Thus the digital footprint slightly beats the bureau score on its own and strongly complements it when both are used together.
- **Main mechanism result.** No single variable drives the result. The explanatory power is spread across several dimensions of the digital footprint, which is exactly what one would expect if it captures multiple facets of borrower type rather than one hidden variable.
- **Main lending-outcome result.** After the firm adopts the digital footprint, default rates drop sharply while access to credit is reshuffled rather than simply shut down. Good-digital-footprint customers gain access; bad-digital-footprint customers lose it.
- **Main financial-inclusion result.** For customers with no credit bureau score, the digital footprint still performs very well. The AUC for unscorable customers is 72.2%, slightly above the 69.6% AUC for scorable customers using the same digital-footprint model.
- **Economic takeaway.** If these results generalize, the information advantage in retail lending may no longer belong only to traditional financial intermediaries or credit bureaus. A digital

footprint can create a low-cost, scalable screening technology that both challenges incumbent lenders and potentially extends credit to some of the unbanked.

2 Data and measurement (key objects)

2.1 Institutional setup

- The data come from a German e-commerce company selling furniture, in a setting similar to Wayfair in the United States.
- Customers register with their name, address, and e-mail and can then choose whether to pay immediately or via invoice after shipment.
- The paper focuses on invoice purchases, because these are economically equivalent to very short-term consumer credit: goods are shipped first and payment is due later.
- Customers have 14 days to pay. If they miss payment, the firm sends three reminders and transfers the claim to a debt-collection agency after about 103 days on average (roughly 3.5 months after the order date).

Interpretation. The financing object in the paper is not a standard installment loan, but it behaves like one economically. The firm is extending unsecured trade credit to consumers and therefore needs a default-screening technology.

2.2 How the firm screens customers

The firm uses two kinds of external / quasi-external information:

- a **basic credit bureau check**, which verifies identity and screens out obviously negative information such as current or recent bankruptcy and apparent fraud;
- a **full credit bureau score**, requested for purchases above EUR 100, which uses credit-history, sociodemographic, and payment-behavior information.

In addition, the firm observes a **digital footprint** for every customer. The digital footprint and the bureau score affect only whether the customer may pay by invoice; they do *not* affect the price of the goods.

The firm combines the full credit bureau score and the digital footprint into a predicted default rate and denies invoice payment if the predicted default rate exceeds 10%.

Interpretation. The paper is about *screening*, not pricing. This matters because differences in default cannot be explained by high-risk customers being charged higher prices or rates.

2.3 Sample construction

- The raw sample contains 270,399 purchases between October 19, 2015 and December 2016.
- The paper restricts attention to purchases for which the firm requested the full credit bureau score, effectively purchases above EUR 100.

- Within this sample, 254,819 observations (about 94%) are **scorable customers** for whom a full credit bureau score exists, and 15,580 observations (about 6%) are **unscorable customers** for whom it does not.

Interpretation. Restricting the analysis to customers who pass the firm’s basic screening makes the exercise harder, not easier. Very bad borrowers are already screened out, so the model must distinguish among customers who are closer to the margin of acceptance.

2.4 Default and related outcome definitions

The main default variable is

$$Default_i = \begin{cases} 1 & \text{if the claim is transferred to a debt-collection agency,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The paper also uses:

- a stricter default definition based on **nonrepayment after collection-agency efforts**,
- **loss given default**, measured as the share of the purchase value not ultimately recovered,
- and, in the adoption analysis, a dummy for whether **invoice payment is offered**.

Interpretation. The main default definition captures short-run repayment behavior, while the stricter definition and loss-given-default exercises check whether the digital footprint predicts more persistent or more economically costly failures.

2.5 The credit bureau score

The credit bureau score is continuous and ranges from 0 (worst) to 100 (best).

For scorable customers:

- mean score = 98.11,
- median score = 98.86,
- 99% of observations lie between 90 and 100.

The score is highly skewed, but even inside this compressed upper tail the paper finds strong predictive power: a score around 98 corresponds to about a 1% default rate, a score around 95 to about a 2% default rate, and a score around 90 to about a 5% default rate.

Interpretation. The bureau score is already informative and has real discriminatory power. So for the digital footprint to add value, it has to clear a nontrivial benchmark.

2.6 Digital-footprint variables

The digital footprint contains simple, low-cost variables available to any online merchant:

- **Device type:** desktop, tablet, mobile, do-not-track setting.

- **Operating system:** Windows, iOS, Android, Macintosh, other.
- **E-mail host:** Gmx, Web, T-Online, Gmail, Yahoo, Hotmail, other.
- **Channel:** paid click / ad, direct, affiliate, organic, other, do-not-track.
- **Checkout time:** evening, night, morning, afternoon.
- **Typography / e-mail structure:** whether the customer name appears in the e-mail address, whether the e-mail contains numbers, whether the customer types name and address in lowercase, and whether the e-mail address contains an error.

The paper's interpretation is that these variables proxy for:

- income / wealth (for example, iOS or premium e-mail hosts),
- character and impulse-control traits (for example, paid-ad entry or nighttime purchases),
- reputation and care (for example, e-mail names, errors, or sloppy text entry).

Interpretation. The striking feature is not technological complexity. It is the opposite: the paper shows that a surprisingly crude digital trace already contains information content comparable to a traditional bureau score.

2.7 Key descriptive facts

For scorable customers, Table 1 reports:

- mean order amount = EUR 317.75,
- mean age = 45.06,
- mean default rate = 0.9%.

For unscorable customers:

- mean order amount = EUR 324.57,
- mean default rate = 2.5%,
- age is observed only for a small subset, but when observed it is lower on average than for scorable customers.

The paper also notes that if invoice payment is offered, more than 80% of customers choose it, and the purchase-completion rate is much higher than when invoice payment is denied.

Interpretation. Unscorable customers are not weird fringe observations in terms of purchase size, but they are much riskier. This is exactly why the financial-inclusion angle is important.

3 Models and empirical specifications (all equations + interpretation)

3.1 Main scoring regressions for scorable customers

Let CBS_i denote the continuous credit bureau score, DF_i the vector of digital-footprint variables, and X_i additional controls such as age, gender, order amount, and item category. The paper estimates logistic regressions of the form

$$\Pr(Default_i = 1) = \Lambda(\alpha + \beta CBS_i), \quad (2)$$

$$\Pr(Default_i = 1) = \Lambda(\alpha + \gamma' DF_i), \quad (3)$$

$$\Pr(Default_i = 1) = \Lambda(\alpha + \beta CBS_i + \gamma' DF_i + \delta' X_i + FE). \quad (4)$$

The paper reports four main versions in Table 4:

- Column 1: credit bureau score only,
- Column 2: digital footprint only,
- Column 3: credit bureau score plus digital footprint,
- Column 4: combined model plus controls and month / region fixed effects.

Interpretation. The first comparison asks whether the digital footprint is informative at all. The second asks the more important question: does it add information *on top of* the bureau score?

3.2 AUC as the metric of discriminatory power

The key summary statistic is the area under the ROC curve (AUC). The paper interprets AUC as the probability of correctly identifying the good case when comparing one random good case and one random bad case.

- 50% = random guessing,
- 100% = perfect classification,
- around 60% is considered useful in information-scarce settings,
- around 70% or more is the target in information-rich settings.

For scorable customers, the main AUCs are:

- credit bureau only: 0.683,
- digital footprint only: 0.696,
- combined model: 0.736,
- combined model + controls / fixed effects: 0.762.

Interpretation. The digital footprint is not merely “significant” in a statistical sense. It is economically meaningful on the same scale lenders actually use to judge model quality.

3.3 Univariate patterns: what the raw data already show

Table 2 provides the simplest raw evidence. For scorable customers:

- the default rate is 2.12% in the lowest credit-score quintile and only 0.39% in the highest quintile;
- mobile orders default at 2.14% versus 0.74% for desktop orders;
- Android users default at 1.79% versus 1.07% for iOS users;
- T-Online users default at 0.51%, while Yahoo users default at 1.96%;
- customers arriving through paid ads default at 1.11%, while affiliate-traffic customers default at 0.64%;
- night purchases default at 1.97%, far above the unconditional average;
- customers with an e-mail error default at 5.09%.

Interpretation. Even before any multivariate modeling, the digital footprint sorts customers into bins with default-rate differences that are economically large.

3.4 Association across variables and why the digital footprint complements the bureau score

Because most digital-footprint variables are categorical, the paper studies associations using Cramér's V .

The main findings are:

- the association between the credit bureau score and individual digital-footprint variables is small, typically between 0.01 and 0.07;
- the main exception is the naturally high association between device type and operating system, about 0.71, which is why the paper combines them in the multivariate specification.

The paper also shows that combining just two footprint variables already gives striking dispersion in default rates. For example, the combination *Mac + T-Online* has a default rate around 0.36%, while *Android + Yahoo* reaches about 4.30%.

Interpretation. Low correlation with the bureau score is exactly what one wants if the digital footprint is to be a complement rather than a substitute. The combined model adds value because it sees something different.

3.5 Out-of-sample and out-of-time validation

To address overfitting, the paper runs:

- **Nx2-fold cross-validation:** repeatedly split the sample into halves, estimate on one half, predict the other, and average the resulting AUCs over 100 iterations;
- **Out-of-sample / out-of-time tests:** estimate the model on early subsamples and evaluate it on later subsamples.

The key results from Table 5 are:

- digital-footprint AUC falls only from 0.696 in-sample to 0.688 out-of-sample and is 0.692 out-of-time;
- combined-model AUC falls from 0.736 to 0.728 out-of-sample and is 0.739 out-of-time.

Interpretation. The predictive link is not a fragile in-sample artifact. The same simple variables continue to work when the model is pushed out of sample and out over time.

3.6 Alternative default definitions and sample splits

Table 6 checks robustness in several directions:

- using a stricter default definition after collection-agency recovery efforts,
- excluding fraud cases,
- predicting loss given default,
- splitting the sample by order size and by gender.

The headline robustness results are:

- stricter-default AUCs: bureau 0.692, digital 0.723, combined 0.757;
- excluding fraud: bureau 0.681, digital 0.691, combined 0.730;
- loss-given-default R^2 : bureau 0.013, digital 0.062, combined 0.069.

Interpretation. The digital footprint predicts not only whether customers pay late, but also whether the loss is more severe. And the result is not a fraud-only story.

3.7 External-validity regression: forecasting future credit-score changes

The paper asks whether the digital footprint is informative beyond this very short-term online-credit setting. The key regression is

$$\Delta CreditScore_{t+1,t} = \beta_0 + \beta_1 \Delta(DF_t, CreditScore_t) + X + \varepsilon, \quad (5)$$

where:

- $\Delta CreditScore_{t+1,t}$ is the future change in the credit bureau score,
- $\Delta(DF_t, CreditScore_t)$ is the difference between the predicted default probability from the digital-footprint model and the predicted default probability from the bureau-score model,
- X contains controls and, in some specifications, fixed effects.

Table 7 shows:

- without controls, a one-percentage-point higher default probability from the digital footprint relative to the bureau score predicts a future bureau-score decline of about 0.76 points (coefficient -75.86);

- after controlling for the initial credit score, the coefficient remains large and significant at about -28.43 .

Interpretation. If the digital footprint says somebody is worse than the current bureau score suggests, the future bureau score tends to move in that direction. This is powerful evidence that the digital footprint is not only fitting noise in one online setting.

3.8 What part of the digital footprint matters?

Table 8 decomposes the AUC contribution of individual variables.

The largest marginal AUC contributions are:

- e-mail host: +2.44 percentage points,
- computer & operating system: +1.71 percentage points,
- e-mail error: +1.78 percentage points,
- lowercase typing: +1.15 percentage points.

The paper also groups variables into broader buckets:

- financially costly / likely income-related variables have stand-alone AUC 61.03% and marginal AUC +2.20 points;
- variables less obviously tied to income have stand-alone AUC 67.35% and marginal AUC +8.52 points.

Interpretation. The digital footprint is not just a noisy income proxy. A large part of its power comes from behavior-like variables that are harder to map directly into traditional financial covariates.

3.9 Adoption design: default rates and access to credit around October 19, 2015

The firm introduced digital-footprint scoring on October 19, 2015. The paper studies what changed around this date using both univariate comparisons and multivariate regressions.

The main difference-style regressions take the form

$$Y_{it} = \alpha + \beta Post_t + \sum_k \theta_k (Post_t \times Category_{ik}) + X'_{it} \delta + FE + \varepsilon_{it}, \quad (6)$$

where Y_{it} is either a default dummy or an invoice-offered dummy, and the categories capture whether the October 19 rule change added both the bureau score and digital footprint or only the digital footprint, as well as credit-score subgroups.

The main univariate results from Table 9 are:

- **Sample 1 (“Score and Digital Footprint Added”):** default falls from 2.54% to 1.19%, while invoice-offered falls from 96.65% to 90.05%;
- **Sample 2 (“Digital Footprint Added”):** default falls from 3.62% to 2.33%, while invoice-offered rises from 39.00% to 40.11%.

Inside Sample 2:

- high-score customers barely change,
- low-score customers see default fall from 6.33% to 3.75%,
- unscorable customers see default fall from 11.65% to 6.44% with invoice access roughly unchanged around 10%.

Table 10 confirms the default-rate drop in multivariate regressions, and a placebo test one year later yields no comparable effect. Table 11 shows the corresponding access-to-credit reshuffling.

Interpretation. The digital footprint does not simply make the lender stingier. Instead it helps the firm reallocate credit toward applicants with favorable digital traces and away from those with unfavorable traces, thereby lowering losses and in some segments even increasing approval rates.

3.10 Unscorable customers

For customers with no full bureau score, the paper estimates

$$\Pr(\text{Default}_i = 1) = \Lambda(\alpha + \gamma' DF_i + \delta' X_i + FE). \quad (7)$$

The key facts are:

- baseline digital-footprint AUC for unscorables = 0.722,
- with additional controls and fixed effects = 0.803,
- out-of-sample AUC remains high at 0.684 in the baseline and 0.659 with controls.

The same broad raw patterns show up for unscorables as for scorable customers:

- mobile customers default at 6.21%,
- Android users at 4.80%,
- customers with their names in the e-mail default at 1.92% versus 3.93% otherwise,
- customers with an e-mail error default at 12.24%.

Interpretation. This is the paper’s strongest financial-inclusion result. The digital footprint remains informative even exactly where the classical bureau-score technology is absent.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no formal IV in the main scoring regressions

The main default regressions are not causal-treatment regressions in the usual sense. The paper is asking a predictive-information question: given the same set of customers, how much can one learn from the bureau score, from the digital footprint, and from both together?

Interpretation. This matters because the right standard is not “is one coefficient causal?” but rather “does the digital footprint contain incremental information that survives out-of-sample and plausibly maps to true credit risk?”

4.2 Why the digital footprint is not just the bureau score in disguise

Two pieces of evidence support complementarity rather than redundancy:

- Cramér’s V between the bureau score and footprint variables is very low, mostly 0.01 to 0.07;
- Figure 4 shows an R^2 of only about 1% between bureau-score percentiles and digital-footprint percentiles.

Interpretation. If the digital footprint were simply reconstructing the bureau score, the combined model would not jump from 0.683 to 0.736 AUC. The paper’s key result is precisely that the two information sets are weakly correlated but jointly powerful.

4.3 Why observables, time, and place do not explain the result

The paper adds age, gender, order amount, item category, month effects, and region fixed effects. The main AUC and coefficient patterns barely change. The footprint is therefore not just a proxy for:

- where the customer lives,
- when the customer purchases,
- what the customer buys,
- or broad demographic controls observed by the bureau.

Interpretation. This is an important credibility step because many digital variables could have been dismissed as merely repackaging obvious observables.

4.4 Out-of-sample stability is part of the identification story

The OOS and OOS-OOT tests matter substantively, not cosmetically. If the footprint only worked because of overfitting or because the coefficients were unstable over time, the economic interpretation would be much weaker.

Interpretation. The fact that the AUC barely declines out of sample means the footprint is picking up stable structure in repayment behavior rather than transient in-sample quirks.

4.5 Why the future-credit-score regression matters

A natural concern is that the digital footprint only predicts short-term e-commerce payment behavior. The future-bureau-score exercise addresses exactly this concern by asking whether the footprint forecasts subsequent changes in a broader, more traditional credit metric.

Interpretation. This is not perfect causal identification, but it is strong external-validity evidence. It shows that the digital footprint sees something that later enters a traditional bureau score.

4.6 What the adoption design can and cannot identify

The adoption analysis is informative because:

- the scoring technology changes sharply on a known date,
- the paper uses narrow windows around the event,
- controls and fixed effects leave the result essentially unchanged,
- a placebo test one year later is null.

But the design is not a perfect difference-in-differences setup because the digital footprint is introduced to all relevant observations on the same date.

Interpretation. The event study is best viewed as evidence on economic outcomes after adoption, not as a fully clean decomposition of each subcomponent of the new score.

4.7 What the paper still cannot fully separate

The paper is deliberately cautious about mechanism. It cannot fully decompose the footprint into:

- financial characteristics that the lender cannot otherwise observe,
- soft information in the classic relationship-lending sense,
- and behavioral traits such as impulsiveness, self-control, or carelessness.

It also notes the Lucas critique: if consumers know these variables are used, they may strategically change their online behavior.

Interpretation. The paper is strongest on *what the footprint predicts* and *how much it adds*; it is weaker on pinning down the exact psychological or organizational channel behind each variable.

5 Tables and figures

The paper's empirical narrative is unusually clean. It starts by showing what the digital footprint looks like and how risky the customers are, then compares predictive power against the credit bureau benchmark, then stress-tests the result out of sample, and finally shows what changed when the firm actually adopted digital-footprint scoring.

5.1 Table 1 and Figure 1: the benchmark problem is real, and the bureau score is already informative

Read:

Table 1
Descriptive statistics

A. Customers with credit bureau score

Variable	Unit	N	Mean	SD	P25	Median	P75
Order and customer							
Order amount	Euro	254,819	317.75	317.10	119.99	218.90	399.98
Gender	Dummy (0=male, 1=female)	254,819	0.66	0.47	0	1	1
Age ^a	Number	254,613	45.06	13.31	34	45	54
Credit bureau score	Number (0=worst, 100=best)	254,819	98.11	2.05	97.58	98.86	99.41
Payment behavior							
Default	Dummy (0/1)	254,819	0.009	0.096	0	0	0

B. Customers without credit bureau score

Variable	Unit	N	Mean	Std.	P25	Median	P75
Order and customer							
Order amount	Euro	15,580	324.57	319.22	119.99	221.60	399.99
Gender	Dummy (0=male, 1=female)	15,580	0.70	0.46	0	1	1
Age ^a	Number	555	38.20	10.46	30	35	46
Credit bureau score	Number (0=worst, 100=best)	15,580	na	na	na	na	na
Payment behavior							
Default	Dummy (0/1)	15,580	0.025	0.156	0	0	0

This table presents summary statistics for the whole sample. The sample period ranges from October 19, 2015 to December 2016. Panel A provides descriptive statistics for customers with credit bureau score. Panel B provides descriptive statistics for customers without credit bureau score. For variable definitions, see Table A1. Missing information on age indicates that the credit bureau does not have information about a customer's age. Observations with nonmissing age in panel B are cases in which the credit bureau has information about the age of the customer, but not enough information to provide a credit score. na, not available.

^aBased on information from the credit bureau.

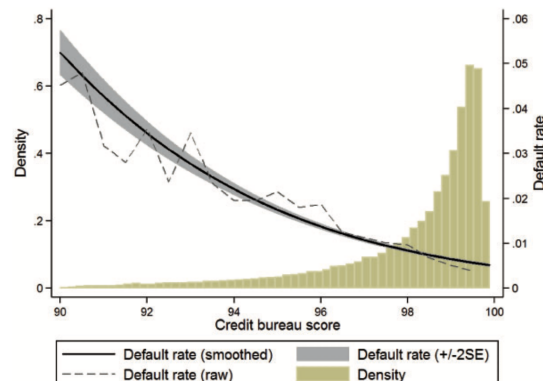


Figure 1
Credit bureau score distribution and default rates

This figure shows the distribution of the credit bureau score and the raw and smoothed default rates as a function of the credit bureau score. *Default(0/1)* is equal to one if the claim has been transferred to a debt collection agency. The smoothed default rates have been determined using a logistic regression and a second-order polynomial of the credit bureau score. The gray-shaded area represents a 2 standard error band around the smoothed default rates using the Delta method. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

- Table 1 shows that scorable customers default at only 0.9%, while unscorable customers default at 2.5%. Mean order amounts are very similar across the two groups.
- Figure 1 shows that the credit bureau score distribution is heavily concentrated near 100, but even within this tight upper tail default rates rise sharply as the score falls.

Economic message. The paper is not comparing the digital footprint to a useless benchmark. The bureau score already has strong discriminatory power, which makes the digital footprint's performance much more impressive.

5.2 Table 2, Table 3, and Figure 2: raw digital-footprint differences are already large

Read:

- Table 2 shows that mobile users, Android users, Yahoo users, nighttime purchasers, lowercase typers, and customers with e-mail errors all default much more often than the baseline categories.
- Table 3 shows that most digital-footprint variables have only weak association with the bureau score, but device and operating system are highly associated with each other, so the paper combines them in the regressions.
- Figure 2 shows that even the combination of two footprint variables — operating system and e-mail host — creates default bins ranging from roughly 0.36% to 4.30%, comparable to or larger than the spread across credit-score deciles.

Economic message. Before any sophisticated estimation, the digital footprint already slices the sample in economically meaningful ways. And because its correlation with the bureau score is so low, these raw differences foreshadow complementarity.

Table 2
Credit bureau score, digital footprint variables, and default rates (scorable customers)

Variable	Value	Observations	Proportion (%)	Default rate (%)	t-test against baseline
Credit bureau score (by quintile)	All	254,819	100	0.04	
	Q1 - lowest	50,969	20	2.12	
	Q2	50,969	20	1.02***	(-14.17)
	Q3	50,969	20	0.91***	(-19.51)
	Q4	51,181	20	0.47***	(-23.87)
Device	Q5 - highest	50,718	20	0.04	(-23.49)
	All	254,819	100	0.94	
	Desktop	145,879	57	0.74	Baseline
	Tablet	45,577	18	0.04	(-21.84)
	Mobile	26,688	11	2.14***	(21.84)
Operating system	Do-not-track setting	36,557	14	0.88***	(2.90)
	All	254,819	100	0.94	
	Windows	114,665	49	0.74	Baseline
	iOS	41,478	16	1.07***	(-0.35)
	Android	11	1	17.99*	(16.46)
E-mail host	Macintosh	21,163	8	0.69	(-6.79)
	Other	1,927	1	1.09*	(-1.74)
	Do-not-track setting	48,577	14	0.88***	(2.66)
	All	254,819	100	0.94	
	Gmail (partly paid)	58,669	23	0.82	Baseline
Channel	Mail (partly paid)	22	0.01	0.04	(-0.70)
	T-Mobile (full service)	30,279	12	0.51***	(-5.32)
	Com (free)	11,625	5	1.06***	(-0.62)
	Yahoo (free, older service)	4,192	2	1.50***	(11.31)
	Hotmail (free, older service)	1,053	5	1.90***	(11.33)
Channel	Other	118,655	49	0.90	(1.38)
	All	254,819	100	0.94	
	Paid	111,399	44	1.11	Baseline
	Direct	45,183	18	0.94	(-4.78)
	Affiliate	24,770	10	0.64***	(-5.68)
Checkout time	Organic	18,290	7	0.69***	(-3.00)
	Other	183,615	7	0.69***	(-6.24)
	Do-not-track setting	36,557	14	0.88	(-3.89)
	All	254,819	100	0.94	
	Evening (6 p.m.-midnight)	108,549	43	0.85	Baseline
Do-not-track setting	Night (midnight to a.m.)	6,913	3	1.19***	(0.49)
	Morning (a.m.-noon)	11,282	6	1.04**	(4.55)
	Afternoon (noon-6 p.m.)	92,756	36	0.89	(0.91)
	All	254,819	100	0.94	
	Yes	218,282	86	0.94	Baseline
Name in e-mail	No	36,557	14	0.88	(-1.12)
	All	254,819	100	0.94	
	Yes	71,017	28	1.24	Baseline
	No	153,802	72	0.82***	(-9.99)
Number in e-mail	All	254,819	100	0.94	
	Yes	213,669	84	0.84	Baseline
	No	41,170	16	1.41***	(10.95)
Is lowercase	All	254,819	100	0.94	
	Yes	225,569	89	0.94	Baseline
	No	19,250	8	2.14***	(18.07)
E-mail error	All	254,819	100	0.94	
	No	251,319	99	0.94	Baseline
	Yes	3,500	1	5.09**	(25.71)

This table provides default rates by credit bureau score quintile and default rates by category of each of the digital footprint variables. The sample is based on scorable customers, that is, the set of customers for which a credit bureau score is available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

Table 3
Correlation/association between credit bureau score, digital footprint, and control variables (scoreable customer)

[illegible]

[†] We exclude customers with a do-not-track setting, as the setting simultaneously applies to device, operating system, and channel information.

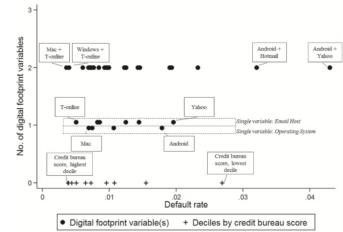


Figure 2
Default rates by combinations of digital footprint variables

This figure shows defaults rates for combinations of the variables *Operating system* and *E-mail host* for all combinations that contain at least 1,000 observations. The x-axis shows default rates, and the y-axis illustrates whether the respect due comes from a single digital footprint variable (e.g., "Android users") or whether it comes from a combination of digital footprint variables (e.g., "Android + Hotmail"). Default rates for credit bureau score deciles are provided as reference points along the row at the very bottom. The sample only includes customers with credit bureau scores. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

5.3 Table 4 and Figures 3–4: the digital footprint matches the bureau score and strongly complements it

Table 4
Default regressions (scorable customers)

	(1)		(2)		(3)		(4)	
	Credit bureau scorepoint		Digital footprint		Credit bureau score & digital footprint		Credit bureau score & digital footprint	
Variables	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Credit bureau score	-0.18*** ^(-3.89)				-0.18*** ^(-4.67)		-0.18*** ^(-4.50)	
Device type & operating system								
Android/Windows			Baseline		Baseline		Baseline	
Android			-0.07 ^(-0.93)		-0.11 ^(-1.03)		-0.19 ^(-1.52)	
Tablet/Android			0.20*** ^(3.19)		0.19*** ^(3.06)		0.33*** ^(4.64)	
Mobile/Android			0.18 ^(1.05)		0.18 ^(1.05)		0.18 ^(1.05)	
Mobile/iOS			0.16*** ^(17.25)		0.15*** ^(15.34)		0.16*** ^(16.71)	
Mobile/Android			0.27*** ^(3.07)		0.27*** ^(3.77)		0.40*** ^(7.26)	
Gen (your party)			0.00		0.00		0.00	
White/party			-0.00		-0.02 ^(-0.22)		-0.01 ^(-0.08)	
T-Office (left column)			-0.06*** ^(-3.90)		-0.05*** ^(-3.35)		-0.07*** ^(-2.47)	
T-Office (right column)			0.14*** ^(3.41)		0.22*** ^(5.06)		0.20*** ^(4.26)	
Y-Office (elder, customer)			0.75*** ^(9.19)		0.77*** ^(8.88)		0.76*** ^(8.22)	
Y-Office (youth, other service)			0.35*** ^(3.77)		0.35*** ^(3.77)		0.35*** ^(3.77)	
Panel (elder, older service)			0.55*** ^(3.77)		0.55*** ^(3.77)		0.55*** ^(3.77)	
Constant								
Observations								
Cluster								
Intercept			-0.48*** ^(-3.35)		-0.48*** ^(-3.58)		-0.48*** ^(-4.31)	
Other			-0.22*** ^(-4.25)		-0.22*** ^(-4.44)		-0.22*** ^(-4.38)	
Intercept			-0.15 ^(-1.79)		-0.15*** ^(-2.04)		-0.15*** ^(-2.04)	
Other			-0.15*** ^(-3.41)		-0.15*** ^(-4.36)		-0.15*** ^(-3.43)	
Missing time								
Evening (6 p.m.-midnight)			0.28*** ^(4.05)		0.28*** ^(4.05)		0.28*** ^(4.79)	
Morning (6 a.m.-noon)			0.42*** ^(4.08)		0.42*** ^(4.08)		0.42*** ^(4.79)	
Afternoon (noon-6 p.m.)			0.78*** ^(7.77)		0.77*** ^(7.77)		0.77*** ^(8.68)	
Night (midnight-6 a.m.)			-0.42*** ^(-6.20)		-0.42*** ^(-6.20)		-0.42*** ^(-6.68)	
Weekend			-0.15*** ^(-3.67)		-0.15*** ^(-3.67)		-0.15*** ^(-3.67)	
Name in e-mail			0.00*** ^(4.50)		0.02 ^(1.31)		0.03*** ^(2.87)	
Phone			0.00*** ^(11.10)		0.00*** ^(11.10)		0.00*** ^(11.10)	
E-mail error			1.06*** ^(20.00)		1.05*** ^(20.36)		1.79*** ^(20.37)	
Constant			-4.89*** ^(-42.87)		-4.89*** ^(-42.87)		-4.89*** ^(-42.87)	
Observations	12,244*** ^(5.76)							
Control for Age, Gender, Education, income, and month	No		No		No		Yes	
Regional fixed effects								
Observations	254,819		254,819		254,819		254,613	
Pseudo R ²	0.24		0.24		0.27		0.26	
LR	0.060		0.060		0.736		0.762	
LR (difference to AIC-BAR)	0.000***		0.000***		0.000***		0.000***	
DIFFERENCE to AIC-BAR	0.186***		0.186***		0.186***		0.267***	
DIFFERENCE to AIC-BAR (1)	0.011*		0.011*		0.020***		0.020***	

We estimate default rate regressions where the dependent variable (*Default97*) is equal to one if the claim has been transferred to a debt collection agency. Column 1 provides results using the credit bureau score as the independent variable; Column 2 provides results using the digital footprint variables as independent variables; Column 3 uses both the credit bureau score and the digital footprint variables as independent variables; and Column 4 adds additional controls (*Age*, *Gender*, *Loan amount*, *Item type*) and month and region fixed effects. All models are estimated using a logistic regression model. Standard errors are adjusted for 97 clusters by 2-digit ZIP codes. The sample is based on scorable customers, that is, the set of customers for which a credit bureau score is available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions,

^a We omit the coefficients for the rare combinations that contain other operating systems, see Table A5 for descriptive statistics.



This figure illustrates the discriminatory power of three different model specifications by providing the receiver operating characteristics curve (ROC curve) and the area under curve (AUC). The ROC curves are estimated using a logit regression of the default dummy on the credit bureau score (light gray), the digital footprint (gray), and both the credit bureau score and a digital footprint (dark gray). The sample only includes customers with credit bureau scores. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

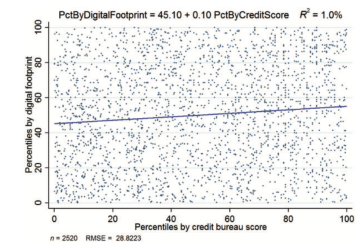


Figure 4
Correlation between digital footprint and credit bureau score (scorable customers)

This figure illustrates the correlation between the credit bureau score and the digital footprint. The x-axis shows percentiles by credit bureau score. The y-axis shows percentiles by the digital footprint. The digital footprint is estimated using the results from Column 2 of Table 4 and multiplied by minus 1 to ensure the same ordering as the credit bureau score (high value = low default probability). The sample only includes customers with credit bureau score and is based on a 1% random sample in order to be able to visualize the results. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

- Figure 4 shows that the bureau score and digital-footprint score are only weakly correlated, with an R^2 of about 1%.

Economic message. This is the core punchline of the paper. The digital footprint is not a cheap imitation of the bureau score. It is a separate information source with similar stand-alone power and large joint value.

5.4 Tables 5 and 6: the performance survives out-of-sample and under tougher outcome definitions

Table 5
Out-of-sample estimates

	(1) Baseline (in-sample)	(2) Out-of-sample	(3) Out-of-sample/out-of-time
AUC credit bureau score	0.683	0.681	0.691
N	254,819	254,819	74,543
AUC digital footprint	0.696	0.688	0.692
N	254,819	254,819	74,543
AUC credit bureau score + Digital footprint	0.736	0.728	0.739
N	254,819	254,819	74,543
AUC credit bureau score + Digital footprint, fixed effects	0.762	0.734	0.730
N	254,613	254,613	74,543

This table provides robustness tests out-of-sample for all main regression specifications. We report AUCs for scorable customers for the model specifications from Table 4. Column 1 reports the baseline results. Column 2 reports out-of-sample estimates of the AUC using Nx2-fold cross-validation. We thereby randomly divide the full sample into half samples A and B. We then estimate a predictive logistic regression using sample A and use the coefficients to create predicted values for observations in sample B. We also estimate a predictive regression using sample B and use the coefficients to create predicted values for observations in sample A. We then determine the AUC for the full sample of observations, using all predicted values estimated out-of-sample. The AUCs reported in Column 2 are the mean AUCs from 100 iterations. In Column 3, we provide out-of-sample/out-of-time estimates. We thereby split the sample in roughly three equally-sized time periods (October 2015–February 2016, March 2016–July 2016, and August 2016–December 2016). The first subperiod is used to estimate the model; the second subperiod is not utilized, to reflect the fact that it takes time to observe the default/no-default outcome; and the third subperiod is used to determine the AUC. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

Table 6
Robustness tests (scorable customers)

A. Default definition	(1)	(2)	(3)	(4)
	Baseline (default = transfer to collection agency)	Default = Write-down	Exclude cases of fraud (9% of defaults)	Loss given default (R^2 reported)
AUC credit bureau score	0.683	0.692	0.681	0.013
AUC Digital footprint	0.696	0.723	0.691	0.062
AUC credit bureau score + digital footprint	0.736	0.757	0.730	0.069
N	254,819	254,819	254,604	2,384
B. Sample splits	(1)	(2)	(3)	(4)
	Small orders < EUR 218.91	Large orders ≥ EUR 218.91	Female	Male
AUC credit bureau score	0.688	0.678	0.689	0.670
AUC Digital footprint	0.711	0.689	0.697	0.700
AUC credit bureau score + digital footprint	0.749	0.729	0.743	0.724
N	127,410	127,409	168,374	86,445

This table provides robustness tests using alternative default definitions as well as various sample splits. Panel A provides results using alternative default definitions. Column 1 reports results using the standard default definition (default = transfer to debt collection agency); Column 2 provides a stricter default definition (default = no full repayment after attempts of debt collection agency). Column 3 excludes fraud cases. Column 4 uses only the sample of defaulted loans and uses the loss given default as the dependent variable. In Column 4 we report the R^2 instead of the AUC, as Column 4 is estimated using a linear regression model while all other models are estimated using a logistic regression. Panel B provides results for various sample splits. All models are estimated using a logistic regression model; apart from Column 4 in panel A, which is estimated using a linear regression model. The sample is based on scorable customers, that is, the set of customers for which a credit bureau score is available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

Read:

- Table 5 shows that out-of-sample and out-of-time AUCs are only slightly lower than the in-sample AUCs.
- Table 6 shows that the digital footprint continues to perform strongly when the default definition is tightened, when fraud cases are excluded, and when the dependent variable is loss given default rather than default itself.

Economic message. These tables matter because they establish that the result is not overfitting and not a fragile artifact of one narrow default definition.

5.5 Tables 7 and 8: the footprint predicts future bureau-score changes and its power is spread across many variables

Read:

- Table 7 shows that when the digital footprint predicts higher default risk than the bureau score, the future bureau score subsequently deteriorates. The relationship survives controls for the initial score plus month and region fixed effects.
- Table 8 shows that no single footprint variable dominates the model. E-mail host, computer

Table 7
Predicting changes in the credit bureau score with the digital footprint

Dependent variable	(1) $\Delta(\text{CreditScore}_{t+1}, \text{CreditScore}_t)$	(2) $\Delta(\text{CreditScore}_{t+1}, \text{CreditScore}_t)$	(3) $\Delta(\text{CreditScore}_{t+1}, \text{CreditScore}_t)$	(4) $\Delta(\text{CreditScore}_{t+1}, \text{CreditScore}_t)$	(5) $\Delta(\text{CreditScore}_{t+1}, \text{CreditScore}_t)$
$\Delta(\text{DigitalFootprint}, \text{CreditBureauScore}_t)$	-25.86*** (-11.86)	-28.42*** (-4.64)	-30.11*** (-5.05)		
Q1 (<100% to <0.49%)				0.88** (2.52)	
Q2 (<0.49% to <0.25%)				0.13* (1.75)	
Q3 (<0.25% to <0.05%)				baseline	
Q4 (<0.05% to <0.15%)				0.08 (0.91)	
Q5 (<0.15% to <100%)				-0.39*** (-3.04)	
DigitalFootprint-Better-Than-CreditBureauScore (D1)					0.33** (2.14)
DigitalFootprint-Better-Than-CreditBureauScore (D1) x					0.80** (2.36)
LowCreditBureauScore					0.03 (0.13)
Q2					baseline
Q3					-0.13 (-0.71)
Q4					0.00 (0.01)
HighCreditBureauScore					FE for each credit score quintile
CreditBureauScore		-0.43*** (-13.47)	-0.42*** (-13.28)	-0.43*** (-10.05)	
Constant	0.37*** (8.75)	-0.98*** (13.51)	absorbed	absorbed	
Month & region fixed effects	No	No	Yes	Yes	
Observations	17,646	17,646	17,646	17,646	17,646
Adj. R ²	.028	.073	.081	.081	.074

This table provides a linear regression of changes in credit bureau scores on the difference between the default probability using the digital footprint and the default probability using the credit bureau score. The independent variable $\Delta(\text{DigitalFootprint}, \text{CreditScore}_t)$ measures the change in credit bureau score between (1) the credit bureau score as of the date of purchase from Table 4 and (2) the latest available credit bureau score up to March 2016. Two credit bureau scores at two different dates are only available when customers return to the E-commerce company at least once between October 2015 and March 2016. Column 2 adds the initial credit bureau score as a control variable; Column 3 adds month and region fixed effects; Column 4 displays results by quintile of $\Delta(\text{DigitalFootprint}, \text{CreditScore}_t)$, and Column 5 tests the predictive power of the digital footprint across credit score quintiles. The dependent variable and the continuous independent variables are winsorized at the 1% and 99% levels. For variable definitions, see Table A1.

Table 8
Marginal AUC for digital footprint variables and combinations of digital footprint variables
A. Individual digital footprint variables (dependent variable: default (0/1))

Variable	Stand-alone AUC (%)	Marginal AUC (PP)
Computer & operating system	59.03	+1.71***
E-mail host	59.78	+2.44***
E-mail Host: paid versus nonpaid dummy	53.80	+0.98***
E-mail Host: Variation within nonpaid e-mail hosts	57.82	+1.79***
Channel	54.95	+0.79***
Checkout time	53.56	+0.63***
Do not track setting	50.40	+0.14*
Name in e-mail	54.61	+0.30**
Number in e-mail	54.15	+0.19**
Is lowercase	54.91	+1.15***
E-mail error	53.08	+1.78***

B. Combinations of digital footprint variables (dependent variable: default (0/1))

Variables	Stand-alone AUC (%)	Marginal AUC (PP)
Potential proxy for income		
Potential proxy for income, financially costly to change (computer & operating system, e-mail host: paid vs. nonpaid dummy)	61.03	+2.20
Unlikely to be a proxy for income, not financially costly to change (nonpaid e-mail host, channel, checkout time, do not track setting, name in e-mail, number in e-mail, is lowercase, e-mail error)	67.35	+8.52
Impact on everyday behavior		
Requires one-time action only (computer & operating system, e-mail host, do not track setting, name in e-mail, number in e-mail)	64.92	+7.25
Requires thinking about how to behave during every individual purchase (channel, checkout time, is lowercase, e-mail error)	62.30	+4.63

This table provides AUCs for each digital footprint variable separately (panel A) and AUCs for selected combinations of digital footprint variables (panel B). The stand-alone AUC is the AUC using only the variable(s) listed in the column "Variable(s)." The marginal AUC of variable(s) X is defined as the AUC of the full model using all digital footprint variables minus the AUC of the model using all variables, except variable(s) X . All models are estimated using a logistic regression model using the default dummy as a dependent variable. The sample is based on scorable customers, that is, the set of customers for which a credit bureau score is available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

/ operating system, lowercase typing, and e-mail errors all contribute materially, and the non-income-style variables jointly matter a great deal.

Economic message. Table 7 gives the result reach beyond one e-commerce environment. Table 8 shows why the footprint is hard to dismiss as “just income” or “just one weird variable.” It is a broad, multi-dimensional signal.

5.6 Figure 5 and Tables 9–11: adoption lowers defaults and reshuffles credit access

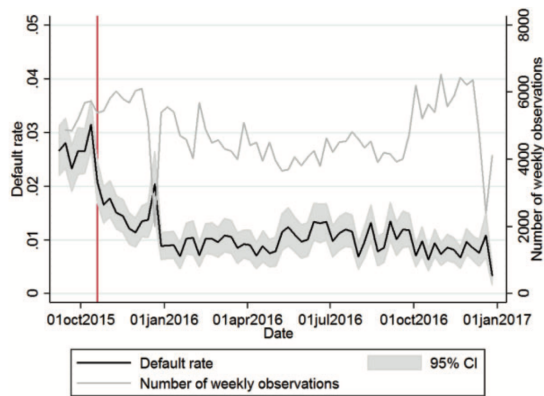


Figure 5
Default rates around the introduction of the digital footprint

This figure illustrates the development of default rates and number of observations around the introduction of the digital footprint. The vertical line indicates October 19, 2015, that is, the date of the introduction of digital footprints. The sample period ranges from September 2015 to December 2016. For variable definitions, see Table A1.

Read:

- Figure 5 shows a sharp drop in default rates around October 19, 2015, the date on which the

Table 9
Development of default rates and access to credit around the introduction of the digital footprint (Univariate results)

	N	Default rate (%)			Invoice offered (%)			Credit bureau score		
		Pre	Post	Δ	Pre	Post	Δ	Pre	Post	Δ
A. Categories										
Sample 1: ScoreAndDFAdded	33,896	2.54	1.19	-1.36***	96.65	90.05	-6.60***	na	98.26	na
Sample 2: DFAdded	10,807	3.62	2.33	-1.29***	39.00	40.11	1.11***	97.82	97.84	0.02
B. Subcategories of "DFAdded"										
DFAdded / High score	3,614	0.84	0.88	0.04	90.00	90.94	0.95	99.42	99.42	0.00
DFAdded / Medium score	4,023	1.82	2.14	0.33	85.21	87.72	2.50***	98.17	98.16	0.00
DFAdded / Low score	2,088	6.33	3.75	-2.57***	31.59	27.52	-4.07***	94.45	94.41	-0.04
DFAdded / Unscorable	1,082	11.65	6.44	-5.22***	10.14	9.59	-0.54	na	na	na

This table depicts default rates, the percentage of customers to which a loan is offered, as well as the average credit bureau score over a ± 6 -week window around the introduction of the digital footprint on October 19, 2015. Panel A depicts averages for two different categories: the category *ScoreAndDFAdded* consists of all observations where the October 19th rule change introduced both a credit bureau score and the digital footprint; the category *DFAdded* consists of all observations where the October 19th rule change introduced the digital footprint but a credit bureau score was requested both prior and after October 19, 2015. Panel B further splits the category *DFAdded* into subcategories *High score* (upper tercile of the credit bureau score distribution, 99.04 < credit bureau score ≤ 100), *Medium score* (middle tercile of the credit bureau score distribution, 96.67 < credit bureau score ≤ 99.05), *Low score* (lower tercile of the credit bureau score distribution, 0 < credit bureau score ≤ 96.67), and *Unscorable* (no credit bureau score available). The number of observations (N) in the second column refers to the number of observations for the default rate analysis. For variable definitions, see Table A1. na, not applicable.

firm introduced digital-footprint scoring.

- Table 9 shows the basic magnitudes: default falls from 2.54% to 1.19% in the sample where both the bureau score and digital footprint are added, and from 3.62% to 2.33% in the sample where only the digital footprint is added on top of the existing bureau-score rule.
- Tables 10 and 11 show that these changes survive multivariate controls, narrower event windows, and placebo tests. For default, the event effect is concentrated among low-score and unscorable customers; for access, the effect looks like reshuffling rather than pure contraction.

Table 10 Development of default rates around the introduction of the digital footprint (multivariate results)						
Dependent variable	(1) Default (0/1)	(2) Default (0/1)	(3) Default (0/1)	(4) Default (0/1)	(5) Default (0/1)	(6) Default (0/1)
Method	Difference post vs. pre	Difference post vs. pre, add categories	Add time trend, controls and FE	Add subgroup	Narrower window around Oct. 19, 2015	Placebo test, 1-year later
Sample	±6 weeks	±6 weeks	±6 weeks	±6 weeks	±4 weeks	±4 weeks
Post	-0.014*** (-9.12)					
Post x ScoreAndDFAdded		-0.014*** (-8.55)	-0.014*** (-8.55)	-0.015*** (-6.13)	-0.015*** (-4.30)	0.001 (0.29)
Post x DFAdded		-0.013*** (-3.85)	-0.012*** (-3.04)			
Post x "DFAdded / High score"				-0.001 (-0.19)	0.000 (0.00)	0.002 (0.78)
Post x "DFAdded / Medium score"				0.005 (0.05)	0.003 (0.03)	0.004 (1.07)
Post x "DFAdded / Low score"				-0.026** (-2.51)	-0.021* (-1.70)	-0.015 (-1.50)
Post x "DFAdded / Unscorable"				-0.050*** (-2.72)	-0.050*** (-2.66)	0.007 (0.43)
Time trend	No	No	0.000 (0.29)	0.001 (0.25)	0.001 (0.15)	-0.002 (-0.30)
Category FE (variables from interaction terms as noninteracted variables)	No	Yes	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes	Yes	Yes
Fixed effects	No	No	Yes	Yes	Yes	Yes
Observations	44,703	44,703	44,703	44,703	30,322	28,905
Adj. R ²	.002	.003	.012	.021	.020	.012

We estimate changes in default rates around the weeks of October 19, 2015. The dependent variable is a default dummy which is equal to one if the claim has been transferred to a debt collection agency. Column 1 provides a regression using a sample window of ±6 weeks around the week of October 19, 2015. Column 2 splits up the post-effect into two categories: the category *ScoreAndDFAdded* consists of all observations where the October 19th rule change introduced both a credit bureau score and the digital footprint; the category *DFAdded* consists of all observations where the October 19th rule change introduced the digital footprint but a credit bureau score was requested both prior and after October 19, 2015. Column 3 adds a time trend (measured in months relative to October 19, 2015, for example, for an order on October 4, 2015, the variable takes on value of 0.50) and controls for order amount and gender and for region fixed effects and fixed effects for the category of the purchased item. Column 4 further splits up the *DFAdded* category into four subcategories, and Column 5 uses a ±4-week window around the week of October 19, 2015. Column 6 provides a placebo test 1 year later. The sample includes both scorable and unscorable customers. All models are estimated using a linear regression model with standard errors clustered by 2-digit ZIP codes. For variable definitions, see Table A1.

Table 11 Development of access to credit around the introduction of the digital footprint (multivariate results)						
Dependent variable	(1) Invoice offered (0/1)	(2) Invoice offered (0/1)	(3) Invoice offered (0/1)	(4) Invoice offered (0/1)	(5) Invoice offered (0/1)	(6) Invoice offered (0/1)
Method	Difference post vs. pre	Difference post vs. pre, add categories	Add time trend, controls and FE	Add subgroup	Narrower window around Oct. 19, 2015	Placebo test, 1-year later
Sample	±6 weeks	±6 weeks	±6 weeks	±6 weeks	±4 weeks	±4 weeks
Post	0.002 (0.30)					
Post x ScoreAndDFAdded		-0.066*** (-25.18)	-0.054*** (-10.31)	-0.046*** (-10.48)	-0.054*** (-9.39)	0.002 (0.25)
Post x DFAdded		0.011* (1.85)	0.011** (2.63)			
Post x "DFAdded / High score"				0.030*** (3.78)	0.008 (0.82)	0.020 (1.62)
Post x "DFAdded / Medium score"				0.044*** (4.37)	0.022* (1.81)	-0.003 (-0.29)
Post x "DFAdded / Low score"				-0.022* (-2.58)	-0.020*** (-2.81)	0.001 (0.09)
Post x "DFAdded / Unscorable"				0.009 (1.57)	0.008 (1.32)	-0.044*** (-3.78)
Time trend	No	No	-0.002** (-2.26)	-0.013*** (-5.40)	-0.002 (-0.33)	-0.014* (-1.91)
Category FE (variables from interaction terms as noninteracted variables)	No	Yes	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes	Yes	Yes
Fixed effects	No	Yes	Yes	Yes	Yes	Yes
Observations	110,438	110,438	110,438	110,438	74,417	65,602
Adj. R ²	.000	.323	.338	.582	.581	.414

We estimate changes in the quantity of lending around the weeks of October 19, 2015. The dependent variable is a dummy of whether or not payment by invoice was offered to a client. Column 1 provides a regression using a sample window of ±6 weeks around the week of October 19, 2015. Column 2 splits up the post-effect into two categories: the category *ScoreAndDFAdded* consists of all observations where the October 19th rule change introduced both a credit bureau score and the digital footprint; the category *DFAdded* consists of all observations where the October 19th rule change introduced the digital footprint but a credit bureau score was requested both prior and after October 19, 2015. Column 3 adds a time trend (measured in months relative to October 19, 2015, for example, for an order on October 4, 2015, the variable takes on value of 0.50) and controls for order amount and gender and for region fixed effects and fixed effects for the category of the purchased item. Column 4 further splits up the *DFAdded* category into four subcategories, and Column 5 uses a ±4-week window around the week of October 19, 2015. Column 6 provides a placebo test 1 year later. The sample includes both scorable and unscorable customers. All models are estimated using a linear regression model with standard errors clustered by 2-digit ZIP codes. For variable definitions, see Table A1.

Economic message. The digital footprint has real operational value. It is not merely a paper prediction exercise. Once adopted, it materially reduces losses and reallocates credit toward customers with better digital traces.

5.7 Tables 12–13 and Figure 6: unscorable customers are where the inclusion value is most visible

Read:

- Table 12 shows that the same basic raw digital-footprint patterns appear among unscorable customers: mobile, Android, e-mail errors, and lack of a name in the e-mail all signal much higher default.
- Table 13 shows that the digital footprint achieves AUC 0.722 for unscorable customers and 0.803 with controls and fixed effects.
- Figure 6 compares the ROC curves for scorable and unscorable customers and shows that the unscorable ROC is at least as strong.

Economic message. This is the broader social implication of the paper. The digital footprint is not just a marginal upgrade for already-scorable customers. It may be a workable screening technology in exactly the segment where traditional bureau information is missing.

6 Short summary

- The paper studies whether a simple online digital footprint can predict consumer default and complement traditional credit bureau information.

Table 12
Digital footprint variables and default rates (unscorable customers)

Variable	Value	Observations	Proportion (%)	Default rate (%)	t-test against baseline
Device	All	15,580	100	2.49	
	Desktop	9,183	59	2.16	Baseline
	Tablet	2,618	17	1.64	(-1.64)
	Mobile	1,546	10	6.21***	(9.07)
	Do-not-track setting	2,233	14	2.28	(0.37)
Operating system	All	15,580	100	2.49	
	Windows	7,763	50	2.19	Baseline
	iOS	2,424	16	2.35	(0.47)
	Android	1,646	11	4.80***	(6.00)
	Macintosh	1,420	9	1.69	(-1.20)
	Other	94	1	7.45***	(3.42)
	Do-not-track setting	2,233	14	2.28	(0.27)
E-mail host	All	15,580	100	2.49	
	Gmx (partly paid)	3,681	24	2.42	Baseline
	Web (partly paid)	3,349	21	2.63	(0.56)
	T-Online (affluent customers)	1,709	11	1.52**	(-2.12)
	Gmail (free)	1,691	11	3.61**	(2.46)
	Yahoo (free, older service)	731	5	3.15	(1.14)
	Hotmail (free, older service)	546	4	2.75	(0.46)
	Other	3,873	25	2.22	(-0.57)
Channel	All	15,580	100	2.49	
	Paid	6,446	41	2.89	Baseline
	Direct	3,257	21	1.87***	(-2.99)
	Affiliate	1,394	9	2.65	(-0.47)
	Organic	1,178	8	2.55	(-0.64)
	Other	1,072	7	2.15	(-1.36)
	Do-not-track setting	2,233	14	2.28	(-1.50)
Checkout time	All	15,580	100	2.49	
	Evening (6 p.m.-midnight)	6,343	41	2.05	Baseline
	Night (midnight-6 a.m.)	369	2	3.52*	(1.91)
	Morning (6 a.m.-noon)	2,959	19	2.74**	(2.08)
	Afternoon (noon-6 p.m.)	5,909	38	2.78***	(2.62)
Do-not-track setting	All	15,580	100	2.49	
	No	13,347	86	2.52	Baseline
	Yes	2,232	14	2.28	(-0.68)
Name in e-mail	All	15,580	100	2.49	
	No	4,432	28	3.93	Baseline
	Yes	11,148	72	1.92***	(-7.26)
Number in e-mail	All	15,580	100	2.49	
	No	12,958	83	1.99	Baseline
	Yes	2,622	17	4.96***	(8.91)
Is lowercase	All	15,580	100	2.49	
	No	14,557	93	2.21	Baseline
	Yes	1,023	7	6.45***	(8.43)
E-mail error	All	15,580	100	2.49	
	No	15,294	98	2.31	Baseline
	Yes	286	2	12.24***	(10.72)

This table provides default rates by category of each of the digital footprint variables. The sample is based on *unscorable* customers, that is, the set of customers for which a credit bureau score is *not* available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

Table 13
Default regressions (unscorable customers)

Variables	(1)		(2)		(3)	
	Digital footprint for unscorable customers		For comparison: Digital footprint for scorable customers (Column 2 of Table 4)		Digital footprint for unscorable customers, fixed effects	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Computer & operating system						
Desktop/Windows	Baseline		Baseline		Baseline	
Desktop/Macintosh	-0.26	(-1.10)	-0.07	(-0.53)	-0.26	(-1.06)
Tablet/Android	-0.22	(-0.86)	0.29***	(3.19)	-0.11	(-0.44)
Tablet/iOS	-0.45*	(-1.72)	0.08	(1.05)	-0.45*	(-1.67)
Mobile/Android	1.07***	(5.97)	1.05***	(17.25)	1.08***	(5.38)
Mobile/iOS	0.63***	(2.69)	0.72***	(9.07)	0.69***	(2.76)
E-mail host ^a						
Gmx	Baseline		Baseline		Baseline	
Web	0.02	(0.11)	0.00	(0.00)	0.01	(0.04)
T-Online	-0.39	(-1.14)	-0.40***	(-3.90)	-0.42	(-1.21)
Gmail	0.33	(1.36)	0.34***	(3.81)	0.31	(1.34)
Yahoo	0.17	(0.61)	0.75***	(9.19)	0.11	(0.36)
Hotmail	-0.02	(-0.06)	0.35***	(3.70)	-0.13	(-0.41)
Channel						
Paid	Baseline		Baseline		Baseline	
Affiliate	-0.08	(-0.39)	-0.49***	(-5.35)	-0.07	(-0.34)
Direct	-0.42**	(-2.34)	-0.27***	(-4.25)	-0.52***	(-2.66)
Organic	-0.05	(-0.24)	-0.15*	(-1.79)	0.03	(0.13)
Other	-0.27	(-1.21)	-0.47***	(-4.50)	-0.18	(-0.82)
Checkout time						
Evening (6 p.m.-midnight)	Baseline		Baseline		Baseline	
Morning (6 a.m.-noon)	0.30*	(1.81)	0.28***	(4.50)	0.32*	(1.88)
Afternoon (noon-6 p.m.)	0.39***	(2.70)	0.08	(1.42)	0.40***	(2.76)
Night (midnight-6 a.m.)	0.44	(1.38)	0.79***	(7.73)	0.45	(1.38)
Do-not-track setting	-0.16	(-0.83)	-0.02	(-0.25)	-0.23	(-1.18)
Name in e-mail	-0.59***	(-4.67)	-0.28***	(-5.67)	-0.54***	(-4.24)
Number in e-mail	0.63***	(4.31)	0.26***	(4.50)	0.61***	(4.07)
Is lowercase	0.95***	(5.45)	0.76***	(13.10)	0.91***	(4.71)
E-mail error	1.66***	(7.81)	1.66***	(20.00)	1.67***	(6.85)
Constant	-3.80***	(-19.20)	-4.92***	(-62.87)	-6.00***	(-11.32)
Control for Gender, Item category, Loan amount, and month and region fixed effects						
	No		No		Yes	
Observations	15,580		254,819		15,580	
Pseudo R ²	.0906		.0524		.1645	
AUC	0.722		0.696		0.803	
(SE)	(0.014)		(0.006)		(0.011)	
Difference to AUC=50%	0.222***		0.196***		0.302***	
AUC (OOS)	0.684		0.688		0.659	

This table provides regression results for a sample of *unscorable* customers utilizing the same model specifications from Table 4. We estimate default rate regressions where the dependent variable (*Default(0/1)*) is equal to one if the claim has been transferred to a debt collection agency. Column 1 provides results using the digital footprint variables as independent variables. Column 2 provides a comparison to the results for the sample of scorable customers. Column 3 adds additional controls (*Gender*, *Loan amount*, *Item type*) and month and region fixed effects. Age is provided by the credit bureau and thus not available for unscorable customers. All models are estimated using a logistic regression model. Standard errors are adjusted for 97 clusters by 2-digit ZIP codes. Out-of-sample AUCs are denoted by AUC (OOS) and are determined using the same methodology from Table 5. The sample in Columns 1 and 3 is based on *unscorable* customers, that is, the set of customers for which a credit bureau score is *not* available. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

We only report coefficients for the 6 largest e-mail providers even though we use the largest 18 categories in the regression (all e-mail providers with at least 1,000 observations). Using only the 6 reported e-mail hosts does not significantly affect the results.

^a We omit the coefficients for the rare combinations that contain other operating systems, see Table A5 for descriptive statistics.

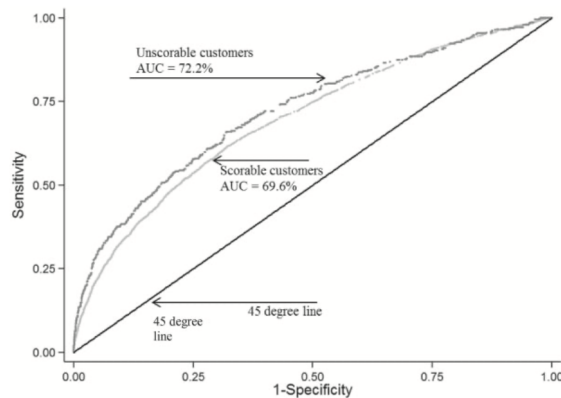


Figure 6
AUC for scorable versus unscorable customers

This figure illustrates the discriminatory power for different samples by providing the receiver operating characteristics curve (ROC curve) and the area under curve (AUC) for scorable customers (light gray) and unscorable customers (dark gray). The ROC curves are estimated using a logistic regression of the default dummy on the digital footprint. The sample period ranges from October 19, 2015 to December 2016. For variable definitions, see Table A1.

- In a German e-commerce setting, the AUC of the digital footprint alone is 69.6%, slightly above the AUC of the bureau score alone (68.3%); the combined model reaches 73.6%.
- The result is robust out of sample, out of time, to alternative default definitions, and to excluding fraud cases.
- The digital footprint also forecasts future changes in the credit bureau score, suggesting that it captures real creditworthiness rather than purely short-run e-commerce repayment behavior.
- When the firm adopts digital-footprint scoring, default rates fall sharply and access to credit is reshuffled toward customers with better digital traces.
- For customers with no bureau score, the digital footprint still performs strongly, with AUC 72.2%, which is why the paper argues that digital footprints may improve financial inclusion.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a very simple digital footprint — made up of device, operating-system, e-mail-host, channel, timing, and typography variables — contains enough information to predict default at least as well as a conventional credit bureau score in this setting. More importantly, the digital footprint adds substantial incremental information when combined with the bureau score.

7.2 Economic intuition

The central intuition is that online behavior leaves behind clues about borrower type that traditional financial data do not fully see. Some of these clues likely proxy for wealth or sophistication, others for self-control or impulsiveness, and still others for reputation and care. Because these dimensions are only weakly correlated with the bureau score, combining the two information sets materially improves screening.

7.3 Takeaway

The deepest takeaway is that the informational frontier in lending may be shifting. In a digital environment, a lender can learn a surprising amount from a borrower's ordinary online trace. This opens two very different possibilities at once: a more competitive and inclusive credit market, and a world in which everyday digital behavior becomes part of the screening technology of finance.